

SENSORS For The AUTOMOBILE INDUSTRY

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Sensors play an important role in the automobile manufacturing industry. In this article we discuss the different sensors that are used in the modern car

Modern cars make thousands of decisions based on the data provided by various sensors that are interfaced to the vehicles' onboard computer systems. A car engine management system consists of a wide range of sensor devices working together, including engine sensors, relays and actuators. Many of these sensors operate in rough and harsh conditions that involve extreme temperatures, vibrations and exposure to environmental contaminants. Yet, these provide vital data parameters to the electronic control unit (ECU) that governs the various engine functions effectively.

In older vehicles, engine sensors and instruments were simple. Modern vehicles are built with complex electronic sensor systems. Digital computers now control engines through various sensors. Luxury cars have a multitude of sensors for controlling various features.

Importance of sensors

Sensors play an important role in automobiles. These enable greater degrees of vehicle automation and futuristic designs. For example, at manufacturing units, sensorised robotic arms are used for painting car bodies and measuring the thickness of the coatings being applied. Manufacturers can simply monitor the thickness of the paint being

sprayed on instruments, airbag claddings and various internal parts of the vehicles using sensors.

Sensors monitor vehicle engines, fuel consumption and emissions, along with aiding and protecting drivers and passengers. These allow car manufacturers to launch cars that are safer, more fuel efficient and comfortable to drive.

Electronic control unit

All sensors inside the vehicle are connected to the ECU, which contains the hardware and software (firmware). Hardware consists of electronic components on a printed circuit board (PCB) with a microcontroller (MCU) chip as the main component. The MCU processes the inputs obtained from various sensors in real time.

All mechanical and pneumatic controls have been replaced by electronic/electrical systems that are more flexible, easier to handle, lighter and cheaper. Moreover, the ECU has reduced the number of wires and emissions, and enabled diagnosing problems with ease. Controlling and monitoring in the modern vehicle is much easier with the ECU.

Communications and control. The ECU simplifies the communication between various components and devices, because long wires for each function are not required. It is installed in the vehicle and connected to the nearest vehicle bus, including controller area network (CAN), local interconnect network (LIN), FlexRay and BroadR-Reach, among others. A CAN bus standard is designed to allow MCUs, sensors and other devices to communicate with each other without a host computer.

Emission control. After sensing fuel level and calculating fuel quantity, the ECU sends signals to various relays and actuators, including ignition circuit, spark plugs, fuel injectors, engine idling

Fig. 1: Some sensors used in cars (Credit: www.digikey.com)

